STANDARD U-100 INSULIN CALCULATOR HELP

The CHLA Insulin Calculator will accurately determine the correct dose of rapid-acting insulin to be administered based on a specific patient order, the grams of carbohydrates eaten, and the measured blood glucose.

Insulin/Carbohydrate Ratio:

- *1 unit of insulin covers **X** grams of carbohydrates eaten*
- Usually determined by 500 / TDD of Insulin
- range: 1 – 100 gm:1 unit; Usual I:C Ratio varies by age
    - <5 yo: 10 – 100
    - 5-11 yo: 4 – 100
    - >=12 yo: 2 – 100

  ✔ ordered value; click to confirm ➔ ✔

  ✖ indicates an ordered value that should be clarified

Correction Factor (also called Insulin Sensitivity Factor):

- *1 unit of insulin reduces a Blood Glucose elevated over Target Blood Glucose by **X** mg/dL*
- Usually determined by 1,800 / TDD of Insulin
- range: 4 – 300; Usual Correction Factor varies by age
    - <5 yo: 50 – 200
    - 5-11 yo: 15 – 200
    - >=12 yo: 4 – 150

  ✔ ordered value; click to confirm ➔ ✔

  ⚠ ordered Correction Factor > 150; uncommon; clarify?

  ✖ ordered value should be clarified

Target Blood Glucose:

- *blood glucose target above which correction must be applied*
- age and individual dependent
- usual target (and range):
    - under 5 yrs: 160 mg/dL (130 – 300 mg/dL)
    - 5 – 12 yrs: 130 mg/dL (120 – 250 mg/dL)
    - over 12 yrs: 130 mg/dL (120 – 250 mg/dL)

  ✔ ordered value; click to confirm ➔ ✔

  ⚠ ordered Target BG is outside the usual range; clarify?

  ✖ ordered value should be clarified

**The calculator is not enabled for input of carbohydrates eaten or measured blood glucose until all above parameters are confirmed by clicking the grey check*

Carbohydrates Eaten:

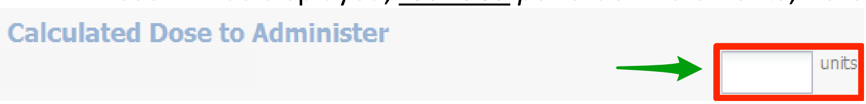
- *Enter the grams of carbohydrates eaten*
- calculator will accept any value from 0 – 300 gm
  - carbs > 250 gm; verify and correct if data entry was incorrect
  - unacceptable entry which must be corrected

Blood Glucose for Calculation:

- *Enter the current glucose measurement*
- calculator will accept any value from 1 – 2,000 mg/dL
  - BG > 600 mg/dL; verify and correct if data entry was incorrect
  - BG > 70 mg/dL: hypoglycemia
    - STOP – Calculator will not calculate dose
    - DO NOT GIVE INSULIN
    - requires rescue intervention for low BG
  - unacceptable entry which must be corrected

To complete calculation of dose to administer:

- *Assure all parameters and patient values are valid*
- Click Calculate button
- Dose will be displayed, ***__rounded__*** *per order increments*, here:



- Transfer calculated dose to MAR after 2RN check and dose has been administered
  - ***__REMEMBER__***
    - *Change time from default to time actually administered*
    - *Change dose from "**1 dose**" to "**X units**" actually administered*